

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SUB-STRAND 2.7 VOLUME AND CAPACITY — FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use what you have learned in all eight lessons about the Kericho tea farmers' three containers: the cylindrical drum (A), the conical sisal cover (B) and the metal frustum bucket (C). Write your answers in the space under each prompt. Use full sentences, show every step of any calculation, include units, and refer to the measurements from the lessons where a question asks you to. Use the formulas and ideas we developed in class, and explain why each one works as well as what it is.

Part 1: The three containers and the 3 : 1 pattern

In Lesson 1 the three Kericho containers were filled to the brim. The drum (A) held about 9.4 L, the conical cover (B) about 3.1 L and the metal bucket (C) about 6.5 L, even though all three have roughly the same height and opening diameter. (a) Work out the ratio of the capacity of A to the capacity of B and say what it suggests. (b) Describe the pouring experiment from Lesson 2 that showed this ratio, and say what you saw. (c) Starting from $V = \pi r^2 h$ for the cylinder, show how you get the volume formula for the cone. (d) Explain in your own words why the cone holds less than the cylinder even though they have the same height and opening width.

Write your answer here:

Part 2: Where does the metal bucket fit?

Container C, the metal bucket, is a frustum of a cone. (a) Where does its capacity (about 6.5 L) sit compared with the cone (about 3.1 L) and the cylinder (about 9.4 L)? (b) Explain how a frustum is related to two cones. (c) Write the volume formula for a frustum of a cone, where R is the top radius, r is the bottom radius and h is the vertical height, and state which two cone volumes are subtracted to obtain it. (d) Explain why the bucket holds more water than the cone even though both have the same height and top opening.

Write your answer here:

Part 3: Pyramids, frustums of pyramids and units

(a) In Lesson 3 you filled a square-based pyramid with rice and poured it into a cuboid with the same base and height. State what you found and write the volume formula for the pyramid. (b) Explain what the cone–cylinder result and the pyramid–cuboid result have in common. (c) Write the volume formula for a frustum of a square pyramid in terms of the areas A_1 and A_2 of its two parallel faces and the height h , and explain what is the same and what is different between this and the frustum of a cone. (d) A container has a volume of $9\,400\text{ cm}^3$. Convert this to litres, and give the conversion relationships you used.

Write your answer here:

Part 4: Designing a container of exact capacity

A Kericho farmer wants a new frustum-shaped metal bucket that holds 10 litres. Available sheet metal fixes the vertical height at $h = 30$ cm and the bottom radius at $r = 8$ cm. (a) Convert the target capacity to cm^3 . (b) Substitute into the frustum formula and write an equation for the top radius R . (c) Solve for R , giving your answer to 1 decimal place, and say whether it is a sensible size for a bucket with $r = 8$ cm. (d) If you built a model bucket like this and found that it held slightly more or less than 10 L when filled with water, give two possible reasons for the difference, based on your project in Lesson 7.

Write your answer here:

Part 5: Answering the driving question

Our driving question was: Why do the tea farmers' containers, a cone, a cylinder and a bucket, look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need? Write a clear explanation of 6 to 8 sentences. Include (i) the relationship between the cylindrical drum and the conical cover, (ii) why the frustum bucket falls between them, (iii) how the formulas let a farmer design a container, and (iv) your answer to the KICD question: why is the knowledge of volume and capacity useful?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of the anchoring phenomenon and data	Uses the Kericho containers and the measured capacities (9.4 L, 3.1 L, 6.5 L) correctly and consistently to support every explanation.	Refers to the containers and most of the data correctly, with minor gaps in linking data to explanations.	Mentions the containers but uses data incorrectly or not at all.
Correct formulas and relationships	States and applies the cone, pyramid, cone-frustum and pyramid-frustum formulas correctly, and explains where the one-third factor and the frustum formula come from.	States the formulas correctly but gives only partial explanation of where they come from.	Formulas are missing, incorrect or only memorised with no reasoning.
Calculation and unit conversion	Calculations are accurate and clearly laid out, with correct conversions between cm^3 , mL and L and correct solving for R.	Method is correct with small arithmetic or unit slips.	Method is unclear or contains major errors in calculation or units.
Scientific explanation of why shapes hold different amounts	Explains clearly, using cross-sections and the cone-removed-from-cone idea, why the cylinder holds most, the cone least and the frustum in between.	Gives a mostly correct explanation, but relies on general statements.	Explanation is incomplete or only describes what the containers look like.
Application, design and communication	Designs a container for a target capacity, identifies realistic sources of error and	Completes the design task and gives some sources of error and uses, with	Design is incomplete and the uses of volume and capacity are unclear.

	explains the usefulness of volume and capacity in clear, well-organised sentences.	minor lapses in clarity.	
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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